Retail Revenue Optimization Report

Executive Summary

This project develops a machine learning model to predict product demand and optimize pricing strategy. By simulating demand across various price points, the analysis identifies the price that maximizes total revenue. The model achieved an R^2 of 0.73 and identified an optimal price of $2.53, resulting in maximum revenue of approximately $4,176.

Business Problem

Retail pricing decisions directly impact revenue and customer demand. Setting prices too high reduces demand, while setting them too low limits revenue potential. This project aims to identify the optimal price point that balances these factors.

Data Overview

The dataset includes:

- Units sold
- Average price
- Revenue
- Time-based features (month, year)
- Engineered demand features (lag and rolling averages)

Methodology

Feature Engineering

- Lag features to capture past demand
- Rolling averages for trend smoothing
- Price change percentage to model elasticity

Model

- Random Forest Regressor
- Predicts units sold based on features

Performance

- MAE: 367.73
- R^2: 0.73

Price Optimization

Demand was simulated across a range of prices using the trained model.

Revenue was calculated as:

- Revenue = Price × Predicted Demand

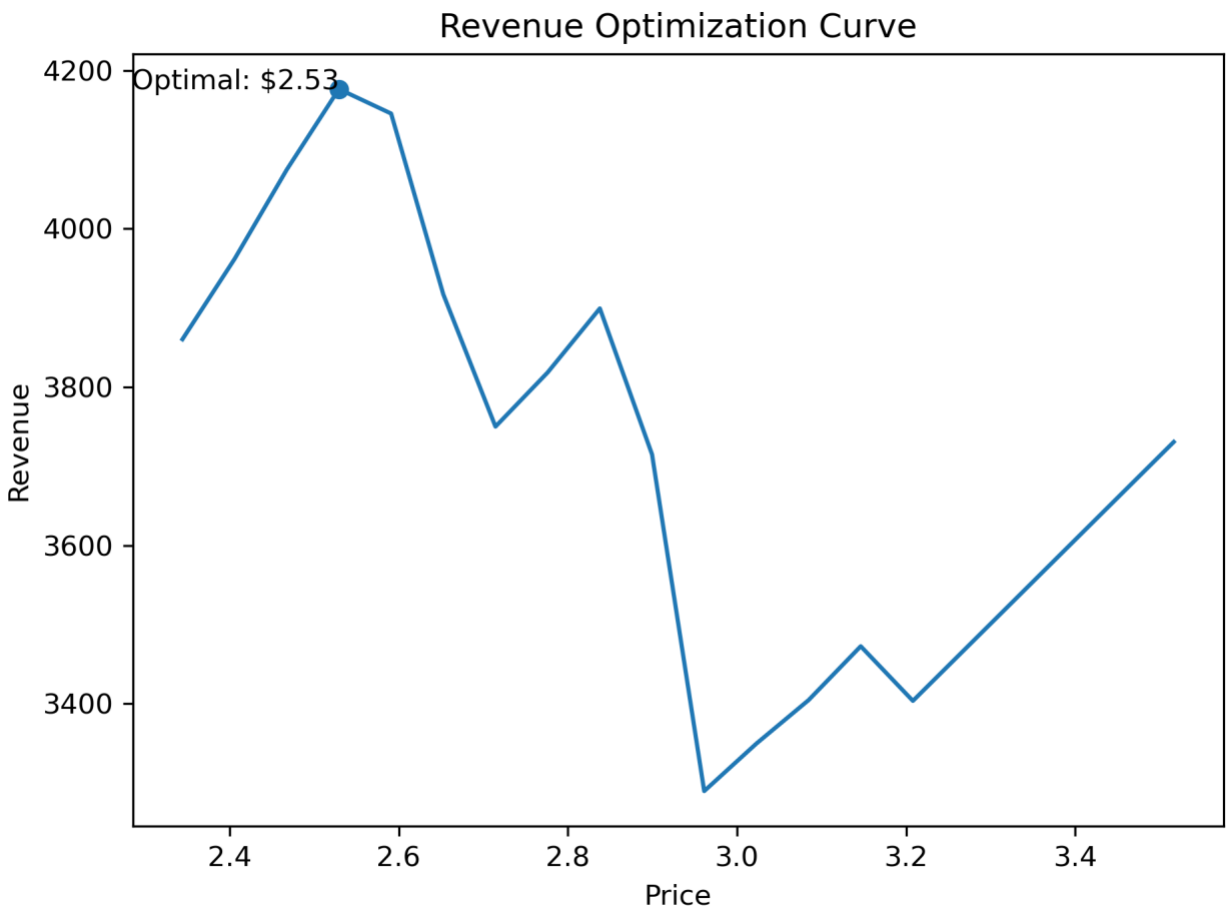
Results

Metric	Value
Optimal Price	$2.53
Predicted Demand	1651
Max Revenue	$4,176

Key Insights

- Demand is highly influenced by price
- Revenue follows a non-linear curve
- A clear optimal price point exists
- Most predictive power comes from a few features

Visualizations



Business Recommendations

- Set price near $2.53 to maximize revenue
- Monitor demand sensitivity to price changes
- Use predictive modeling for pricing decisions

Limitations

- No cost data → cannot optimize profit
- Model assumes historical patterns continue
- Single-product focus

Conclusion

This project demonstrates how machine learning can be applied to pricing strategy. By predicting demand and simulating pricing scenarios, businesses can make data-driven decisions that maximize revenue.